$$
\begin{aligned}
d &= d_1 + d_2 \\
&= \sqrt{2Rh_1} + \sqrt{2Rh_2} \\
&= \sqrt{2R}\left(\sqrt{h_1} + \sqrt{h_2}\right) \\
&= \sqrt{2\times 6.4\times 10^6}\times\left(\sqrt{40} + \sqrt{30}\right) \\
&= 16\times 10^2\sqrt{5}\times(6.32 + 5.48) \\
&= 42217\,\text{m} = 42.217\,\text{km}
\end{aligned}
$$

10.12 SOME IMPORTANT COMMUNICATION SYSTEMS

There are a large number of communication systems available in the market for varied purposes. Here some of the important communication systems are introduced and their applications are briefly discussed.

10.12.1 SATELLITE AND ITS COMMUNICATION

The satellite communication is a mode of transmission of the signal between transmitter and receiver via satellite. The message signal from the Earth station is transmitted to the satellite on board via an uplink (frequency band 6 GHz), amplified by a transponder and then retransmitted to another Earth station via a downlink (frequency band 4 GHz) (Figure 10.55).

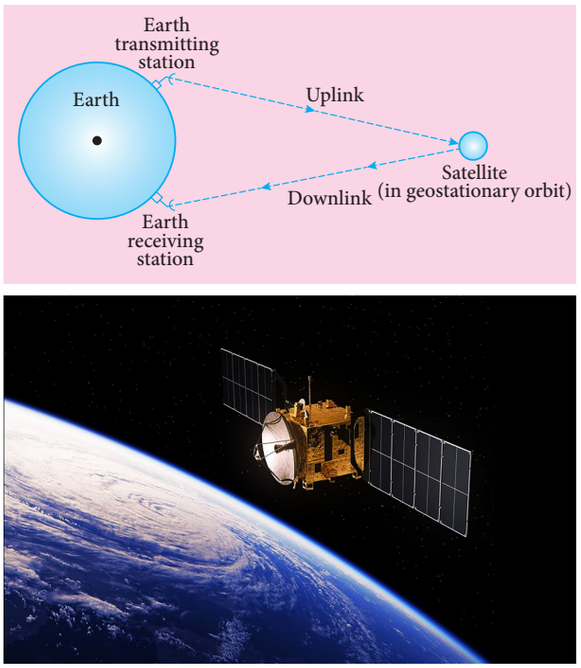


Figure 10.55 Satellite communication system

Applications

Satellites are classified into different types based on their applications.

i) **Weather satellites:** They are used to monitor the weather and climate of Earth. By measuring cloud mass, these satellites enable us to predict rain and dangerous storms like hurricanes, cyclones etc.

ii) **Communication satellites:** They are used to transmit television, radio, internet signals etc. Multiple satellites are used for long distance communication.

iii) **Navigation satellites:** These are employed to determine the geographic location of ships, aircrafts or any other object.

10.12.2 FIBRE OPTIC COMMUNICATION

The method of transmitting information from one place to another in terms of light pulses through an optical fiber is called fiber optic communication. It works on the principle of total internal reflection.

XII_Physics_Vol-2_UNIT-10_Part 2_EM.indd 240 04-03-2022 17:19:25